

grids, industrial automation, and healthcare monitoring.

It supports open data formats like RestAPI, JSON, XML and Java.

Integration: A middleware platform should have no vendor lock-in to ensure seamless integration of enterprise-wide tools, applications, products and systems developed/deployed by different organisations and vendors. It should support open APIs and the deployment model of the cloud, and be highly available.

Reusability: Middleware makes designing and development easier by modifying system components and assets for specific requirements, leading to cost efficiency.

Maintainability: Middleware helps maintain the network efficiently.

Cost: Using middleware can reduce costs by minimising the need for custom integration, and allowing organisations to leverage existing technologies and protocols.

Device management: Middleware provides tools and frameworks for managing the lifecycle of IoT devices, including provisioning, configuration,

monitoring, and maintenance.

Data management: Middleware facilitates the efficient collection, storage, processing, and retrieval of data from IoT devices. This ensures that data is accessible, organised, and ready for analysis and decision-making.

Security: IoT middleware includes security features to protect data and devices from unauthorised access and cyber threats. By implementing robust security measures, middleware helps safeguard sensitive information and maintain the integrity of IoT systems. It supports security features like encryption, authentication, authorisation, and auditing.

Next-generation technologies: IoT middleware supports technologies such as M2M applications, real-time analytics, machine learning, artificial intelligence, and event reporting.

Components of IoT middleware architecture

IoT middleware architecture must ensure that IoT systems are scalable, secure, and efficient, enabling seamless communication and data exchange

between various components.

The middleware should support any data-producing entity as well as any data-consuming entity. Access to the middleware should be enabled via well-designed and documented APIs. These APIs should support application protocols like HTTPS, AMQP, MQTT, WebSocket, etc, and incorporate standard security schemes.

The owner of the middleware should be able to create accounts, provide identity credentials, and define access rights to the users of the middleware. Authenticated and authorised users/applications should be able to access live and archival data related to a sensor/device. Applications should be able to upload/publish data to the middleware, provided they are authenticated and authorised. Middleware should be agnostic to specific implementation choices for the components but be able to incorporate emerging tech advances.

The components/layers of the IoT middleware architecture are:

Device layer: This includes sensors, actuators, and other IoT devices that collect and transmit data.

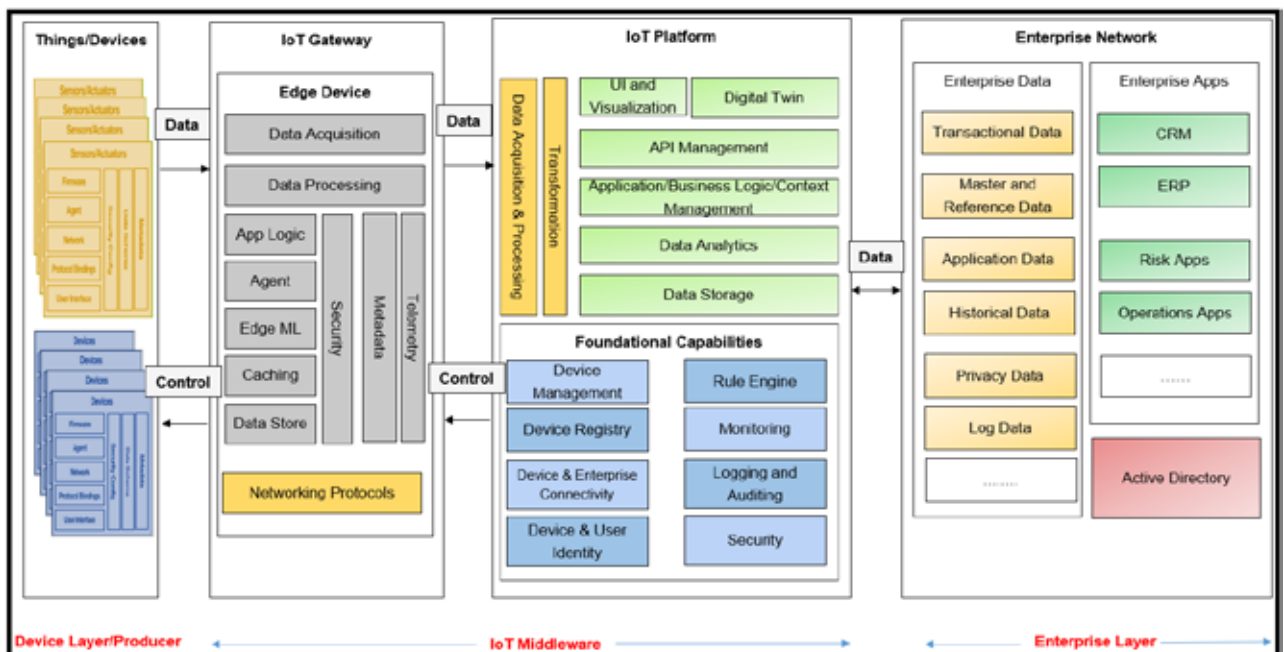


Figure 1: IoT middleware architecture